

Logarithmic Functions & Properties

ANSWER KEY



Converting between exponential and logarithmic form

- 1 Rewrite $2^5 = 32$ in logarithmic form, and $\log_3 81 = 4$ in exponential form.
 $2^5 = 32 \Leftrightarrow \log_2 32 = 5$. $\log_3 81 = 4 \Leftrightarrow 3^4 = 81$.
- 2 Evaluate $\log_5 125$ and $\log_2 \frac{1}{8}$ without a calculator.
 $\log_5 125 = 3$ since $5^3 = 125$. $\log_2 \frac{1}{8} = -3$ since $2^{-3} = \frac{1}{8}$.

Properties of logarithms

- 3 Use logarithm properties to expand $\log \left(\frac{x^3 \sqrt{y}}{z^2} \right)$ fully.
 $\log \left(\frac{x^3 \sqrt{y}}{z^2} \right) = \log(x^3) + \log(y^{1/2}) - \log(z^2) = 3\log x + \frac{1}{2}\log y - 2\log z$.
- 4 Combine $2\log x - \frac{1}{3}\log y + \log z$ into a single logarithm.
 $2\log x - \frac{1}{3}\log y + \log z = \log(x^2) - \log(y^{1/3}) + \log z = \log \left(\frac{x^2 z}{y^{1/3}} \right)$.
- 5 Use the change of base formula to evaluate $\log_7 50$, to three decimal places.
 $\log_7 50 = \frac{\ln 50}{\ln 7} = \frac{3.912}{1.946} \approx 2.010$.

Visualizing logarithmic functions

- 6 The graph shows $f(x) = \ln(x)$, the natural logarithm.
 - a State the domain and the vertical asymptote shown in the graph. Domain: $x > 0$. Vertical asymptote at $x = 0$, where the curve plunges toward $-\infty$ as $x \rightarrow 0^+$.
 - b State the x-intercept shown. x-intercept at $(1, 0)$, since $\ln(1) = 0$.

Solving exponential equations using logarithms

- 7 Solve $5^x = 40$ for x , to three decimal places.
Taking log of both sides: $x\log 5 = \log 40$, so $x = \frac{\log 40}{\log 5} \approx \frac{1.602}{0.699} \approx 2.292$.
- 8 Solve $3(2)^{x-1} = 45$ for x , to three decimal places.
 $2^{x-1} = 15$. Taking log: $(x-1)\log 2 = \log 15$, so $x-1 = \frac{\log 15}{\log 2} \approx 3.907$, giving $x \approx 4.907$.

Solving logarithmic equations

- 9 Solve $\log_2(x + 3) = 4$ for x , and check that the solution keeps the argument positive.
 $x + 3 = 2^4 = 16$, so $x = 13$. Check: $x + 3 = 16 > 0$ ✓. Solution: $x = 13$.
- 10 Solve $\log(x) + \log(x - 3) = 1$ for x , checking both solutions against the domain.
 Combine: $\log[x(x - 3)] = 1$, so $x(x - 3) = 10$: $x^2 - 3x - 10 = 0$, $(x - 5)(x + 2) = 0$: $x = 5$ or $x = -2$.
 Since $x > 0$ and $x - 3 > 0$ are required (domain of the original logs), $x = -2$ is rejected. Solution: $x = 5$.

Applications: the pH scale and decibels

- 11 The pH of a solution is given by $\text{pH} = -\log[H^+]$, where $[H^+]$ is the hydrogen ion concentration in mol/L. Find the pH of a solution with $[H^+] = 3.2 \times 10^{-5}$ mol/L, to two decimal places.
 $\text{pH} = -\log(3.2 \times 10^{-5}) \approx -(-4.495) = 4.49$.
- 12 Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. If one sound is 1000 times more intense than another, find the difference in their decibel levels.
 Difference $= 10\log(1000) - 10\log(1) = 10\log(1000) = 10(3) = 30$ decibels.

Transformations of logarithmic functions

- 13 Describe how the graph of $g(x) = \ln(x + 2) - 3$ is transformed from the parent function $f(x) = \ln(x)$, and state its new vertical asymptote and domain.
 $g(x) = \ln(x + 2) - 3$ shifts $f(x) = \ln(x)$ left 2 units and down 3 units. The vertical asymptote shifts from $x = 0$ to $x = -2$, and the domain becomes $x > -2$.
- 14 Find the domain of $h(x) = \log(4 - x^2)$.
 Require $4 - x^2 > 0$: $x^2 < 4$, so $-2 < x < 2$. Domain: $(-2, 2)$.

Solving equations that combine exponentials and logarithms

- 15 Solve $e^{2x} - 5e^x + 6 = 0$ by substituting $u = e^x$.
 Let $u = e^x$: $u^2 - 5u + 6 = 0$, so $(u - 2)(u - 3) = 0$, giving $u = 2$ or $u = 3$. Since $u = e^x > 0$ for all real x , both are valid: $e^x = 2 \Rightarrow x = \ln 2$, or $e^x = 3 \Rightarrow x = \ln 3$.
- 16 Solve $\log_2(x) + \log_2(x - 2) = 3$ for x , checking the domain.
 Combine: $\log_2[x(x - 2)] = 3$, so $x(x - 2) = 2^3 = 8$: $x^2 - 2x - 8 = 0$, $(x - 4)(x + 2) = 0$: $x = 4$ or $x = -2$. Domain requires $x > 0$ and $x - 2 > 0$ (i.e. $x > 2$), so $x = -2$ is rejected. Solution: $x = 4$.

Comparing logarithmic growth to other function families

- 17** Explain why logarithmic functions grow much more slowly than linear functions for large x , and give a numerical comparison of $\log(x)$ and x at $x = 1000$.

Logarithms compress large inputs: each tenfold increase in x adds only 1 to $\log(x)$, while a linear function adds proportionally to the increase in x itself. At $x = 1000$: $\log(1000) = 3$, while the linear function $y = x$ gives 1000 — the logarithm is vastly smaller, illustrating its extremely slow long-run growth.

Modeling with logarithms: the Richter scale

- 18** Earthquake magnitude on the Richter scale is $M = \log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. An earthquake measures magnitude 7, while another measures magnitude 5. Find how many times more intense the magnitude-7 earthquake is.

$7 - 5 = \log\left(\frac{I_7}{I_0}\right) - \log\left(\frac{I_5}{I_0}\right) = \log\left(\frac{I_7}{I_5}\right) = 2$, so $\frac{I_7}{I_5} = 10^2 = 100$. The magnitude-7 earthquake is 100 times more intense.

- 19** If an earthquake's intensity is 10,000 times a reference intensity I_0 , find its Richter magnitude.

$$M = \log(10000) = \log(10^4) = 4.$$